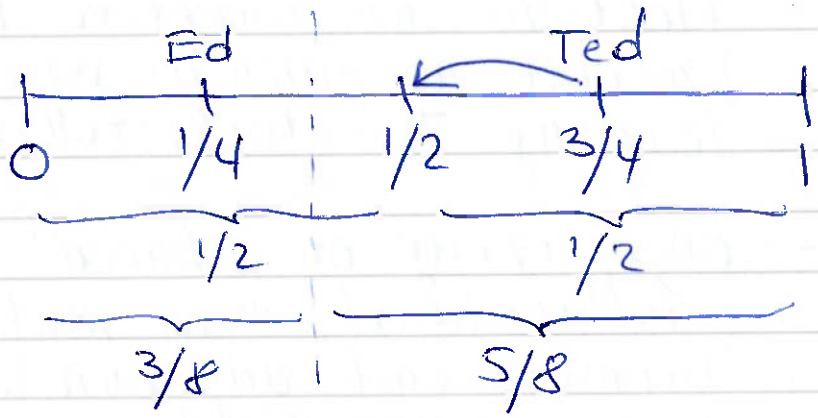
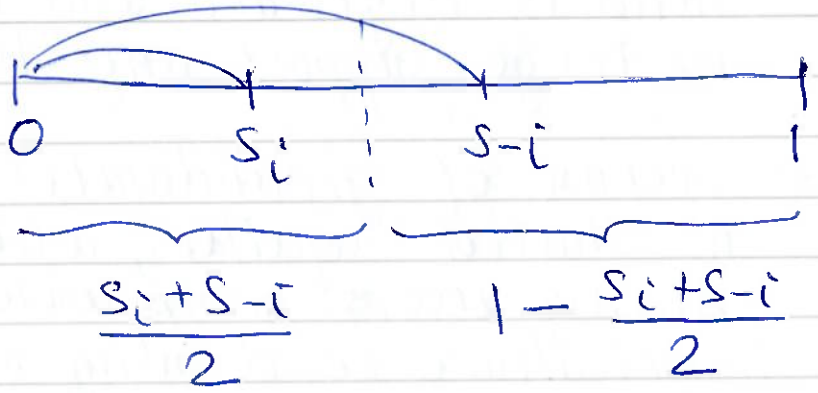
week 1 (19.01.2026)

(0) module admin + coursework

(1) The Ice-Cream wars





(2) Prisoners' dilemma

[A] Youtube video

B

Tragedy of the commons: examples

- overfishing in international waters, where no single country owns the fish stocks, so each fleet has an incentive to harvest as much as possible, leading to stock collapse.

- overgazing on shared pástureland, when multiple farmers each add more animals to a common field until the land is degraded and can no longer support any herd.

- overuse of groundwater from a shared áquifer, where many users pump water for agriculture or drinking without coordination, eventually depleting or polluting the source.

- Air pollution and greenhouse-gas emissions where firms and individuals benefit privately from emitting while the costs (climate change, health effects) are borne collectively.

- Deforestation of open-access forests, when logging and land clearing by many actors for short-term gain destroy biodiversity and long-run ecosystem services.

Examples in MAS & AI (exercise)

In multiagent systems and AI, tragedy-of-the-commons situations appear whenever many learning or planning agents share a limited resource and optimise only their own reward:

- shared communication or computation bandwidth: multiple agents send messages over a common channel or use a shared computer cluster; if each agent transmits or scheduled jobs agressively, congestion and packet loss or long queues reduce performance for all.
- autonomous vehicles in traffic networks: self-driving cars that individually choose time- and route-optimal policies can all flock to the same "fast" route, creating traffic jams that slow everyone down.
- open multiagent platforms and reputation systems: in open generative multiagent systems, agents might overuse shared content, models or public goods (e.g., exploiting a shared reputation mechanism or

public knowledge base) in ways that degrade its quality.

Origin (start with it)

The phrase "tragedy of the commons" comes from the title of a 1968 essay by ecologist Garrett Hardin, published in the journal Science. Hardin borrowed and popularized an earlier idea from the 19th-century writer William Forster Lloyd, who described overgazing on shared pastureland ("the commons") as a way to explain how individually rational use of a shared resource can lead to collective ruin.

③ Beauty contest

Pollevery where

+ discussion

4. The Professor's dilemma
   find NE
5. The Ice Cream wars
   find NE
6. Battle of Sexes
   find NE
7. Exercise 1
8. Exercise 2
9. Exercise 3

Poll everywhere
+
discussion

(10) Battle of Sexes

[A] Find PSNE and SE (in the original game)

[B] Find mixed NE in the original and modified game

[C] Find maximin and minimax (pure and mixed) (in the modified game)

[A]

	q	$1-q$
p	2,1	0,0
$1-p$	0,0	1,2

PSNE & SE

[B]

$p = 2(1-p)$

$p = 2 - 2p$

$3p = 2 \Rightarrow \boxed{p = \frac{2}{3}}$

$2q = 1 - q$

$3q = 1 \Rightarrow \boxed{q = \frac{1}{3}}$

	q	$1-q$
p	2,1	0,0
$1-p$	0,0	1,3

$p = 3(1-p)$

$p = 3 - 3p$

$4p = 3 \Rightarrow \boxed{p = \frac{3}{4}}$

$2q = 1 - q$

$3q = 1 \Rightarrow \boxed{q = \frac{1}{3}}$

$\boxed{C}$

			P1 minimax
2,1	0,0	0	1
0,0	1,3	0	3
P2 maximin → 0	0	↑ P1 maximin	
2	1 ← P2 minimax		

	q	$1-q$
p	2,-2	0,0
$1-p$	0,0	1,-1

$$-2p = -(1-p)$$
$$-2p = -1+p$$
$$1 = 3p \Rightarrow \boxed{p = \frac{1}{3}}$$

$$\boxed{u1 = \frac{2}{3}}$$ P1 maximin

$$2q = 1-q$$
$$3q = 1 \Rightarrow \boxed{q = \frac{1}{3}}$$ P2 minimax

	q	$1-q$
p	$-1, 1$	$0, 0$
$1-p$	$0, 0$	$-3, 3$

$$p = 3(1-p)$$
$$p = 3 - 3p$$
$$4p = 3 \Rightarrow \boxed{p = \frac{3}{4}}$$ P1 minimax

$$\boxed{U2 = \frac{3}{4}}$$

$$-q = -3(1-q)$$
$$-q = -3 + 3q$$
$$3 = 4q \Rightarrow \boxed{q = \frac{3}{4}}$$ P2 maximin

Compare to pure maximin and minimax, conclude that mixed strategies are more effective